



Cultivating change: A study of Western Canadian dairy farmers' perspectives on mastitis-related antimicrobial stewardship and strategies for milk culturing adoption

Jennifer A. Ida,¹ Ellen de Jong,² Anne-Marieke C. Smid,^{2,3} Kristen K. Reyher,⁴ David F. Kelton,⁵ Caroline Ritter,⁶ Inge van der Velden,^{2*} and Herman W. Barkema^{2†}

¹Department of Veterinary and Biomedical Sciences, The Pennsylvania State University, State College, PA 16802

²Faculty of Veterinary Medicine, University of Calgary, Calgary, AB, Canada T2N 4N1

³Wageningen Livestock Research, Wageningen University & Research, 6708 WD, Wageningen, the Netherlands

⁴University of Bristol Veterinary School, Langford, Bristol BS40 5DU, United Kingdom

⁵Department of Population Medicine, Ontario Veterinary College, University of Guelph, Guelph, ON, Canada N1G 2W1

⁶Department of Health Management, University of Prince Edward Island, Charlottetown, PE, Canada C1A 4P3

ABSTRACT

Mastitis prevention and treatment decisions have been the focus of antimicrobial stewardship (AMS) in the dairy sector, as antimicrobials used for prevention and treatment of mastitis cases account for ~65% of antimicrobials used on dairy farms. To improve on-farm AMS, a selective treatment approach to clinical mastitis (CM) is promoted, guided by pathogen identification through diagnostic techniques, such as milk culturing. To design effective interventions to support the adoption of pathogen identification, it is important to understand dairy farmers' perceptions regarding mastitis-related AMS and identify their barriers and motivators around the adoption of milk culturing as part of their CM treatment protocol. As such, interviews with 17 dairy farmers or farm personnel located in Western Canada were conducted and analyzed both inductively using thematic analysis and deductively using the COM-B (capability, opportunity, and motivation) framework, a part of the broader Behavior Change Wheel. The latter is a framework designed to assess factors that determine behavior and aid in the development of relevant interventions and was used to design potential interventions to mitigate the identified barriers and promote the facilitators (e.g., awareness that culturing is valuable, support of herd veterinarian) that were identified in the interviews. Four themes were identified regarding AMS stewardship decisions: (1) responsible in principle, unintentionally variable in practice; (2) upstream solutions: prevention, herd health, and good management as foundations of judicious AMS;

(3) the need for consistent engagement of the veterinarian as an external advisor on AMS; and (4) behavioral change around AMS: generational differences, internal drive, external support. Based on the interview results, 10 interventions were suggested, including encouragement of milk culturing through veterinarians and farm magazines; training to improve aseptic sampling; education to improve interpretation of test results and increase problem awareness; offering comprehensive udder health packages that include milk sampling to enable farmers to send in samples; increasing familiarity of culturing through on-farm demonstrations and among peer groups; and incentivizing farmers to adopt milk culturing through benchmarking. Implementing these interventions could lead to increased uptake of selective CM treatment protocols, thereby contributing to AMS on dairy farms.

Key words: antimicrobial stewardship, clinical mastitis, behavioral framework, motivator, interventions

INTRODUCTION

Clinical mastitis (CM) treatment decisions have been the focus of antimicrobial stewardship (AMS) in the dairy sector, as the prevention and treatment of mastitis accounts for 60% to 70% of antimicrobials used on dairy farms (Pol and Ruegg, 2007; Menéndez González et al., 2010; Kuipers et al., 2016; Tommasoni et al., 2023; Warder et al., 2023). Clinical mastitis treatments are commonly administered based on clinical signs and, in some cases, other diagnostics (e.g., California Mastitis Test), without use of pathogen identification diagnostics (Donadeu et al., 2020). However, in some nonsevere CM cases, treatment with an antimicrobial may not be necessary or beneficial. This includes nonsevere CM cases in which the cause is gram-negative bacteria or cases in which no bacterial growth is present on bacterial

Received November 4, 2025.

Accepted March 12, 2026.

*Deceased.

†Corresponding author: barkema@ucalgary.ca

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

culturing of the milk. To improve mastitis-related AMS, a selective treatment approach to CM is advised, where decisions center around pathogen identification using techniques such as milk culturing and result in only treating those cases caused by gram-positive bacteria, such as staphylococci and streptococci (de Jong et al., 2023a). Although this approach has proven to result in similar treatment outcomes (e.g., resolution of clinical signs, return to prior milk production, return to low SCC) compared with a protocol where all CM cases are treated with antimicrobials (de Jong et al., 2023b), uptake of such protocols based on milk culture results among Canadian farmers is still relatively low (de Jong et al., 2023a,b).

Recent evidence underscores that critically important antimicrobials for human health may not be essential for the treatment of nonsevere CM cases (Nobrega et al., 2020). European countries that have introduced regulations limiting the use of critically important antimicrobials for the treatment of dairy cattle in the absence of diagnostic and susceptibility testing have achieved notable success in reducing the use of these targeted critical antimicrobials (Armstrong et al., 2018; Turner et al., 2018). Farmers in Belgium, the UK, and the Netherlands have pursued reduction in antimicrobials with no reported negative impacts on animal health and welfare (Armstrong et al., 2018; Turner et al., 2018). As of December 2018, all medically important antimicrobials (MIA) have been strictly regulated across Canada, requiring an active veterinarian–client–patient relationship and veterinary prescription (Health Canada, 2024). In Quebec, category I antimicrobials (e.g., ceftiofur, polymyxin B sulfate), which includes the majority of available CM treatments, have been prohibited as a first line of treatment without veterinary consultation since 2019 (Roy et al., 2020).

In Canada, dairy farmers have several options to test milk samples from cows for the presence of bacteria. Veterinary practices may offer diagnostic testing, including culturing of milk samples. Independent laboratories also provide milk culture services. Turnaround times vary, typically ranging between 1 and 4 d, with limited availability on weekends and holidays. Farmers are informed of results through texts, emails, or web notifications. Lactanet (Guelph, ON, Canada), the Canadian DHI organization, offers a PCR test of 4 pathogens based on individual cow samples taken during a routine DHI visit, with sampling frequency ranging from 4 to 10 times per year on individual farms. Farmers also have the option to purchase on-farm milk culturing systems, although the sensitivity of these on-farm tests to identify gram-positive bacteria ranges from 59% to 98%, and specificity ranges from 48% to 97% (Malcata et al., 2020).

Effectively promoting AMS through novel techniques and practices, such as selective treatment of CM on dairy farms, requires a holistic understanding of the breadth

of factors (e.g., personal values, underlying beliefs, social relationships) and contextual influences that shape farmers' decisions and behaviors (Garforth, 2015). This includes understanding how multiple factors, including but not limited to financial considerations, interact with one another to influence farmers' decisions. A growing body of research has been conducted to understand determinants of dairy farmers' behavior in the context of antimicrobial use (AMU), summarized by Farrell et al. (2021). These determinants include farmers' knowledge and awareness of antimicrobial resistance (AMR; Raymond et al., 2006; Jones et al., 2015; Rees et al., 2021a), experience and ability to accurately judge animal health (Swinkels et al., 2015; McDougall et al., 2017; Rees et al., 2021b), availability of individual cow data and records (Brunt et al., 2025), relationship with their veterinarian (Fischer et al., 2019), drug attributes (Redding et al., 2014), farm economics (Jones et al., 2015; Golding et al., 2019), cow characteristics (de Jong et al., 2024a; Vaarst et al., 2002), and recommendations of other farmers (Swinkels et al., 2015).

Although recent intervention studies have been informed by theories such as self-determination theory (Rees et al., 2021a), most research focused on identifying the motivators and barriers influencing farmers' behavior and decision-making processes have done so without grounding these studies in a theoretical framework (reviewed by Biesheuvel et al., 2021). Providing a theoretical underpinning may contribute to improved uptake of recommended practices. The behavior change wheel (BCW), developed by Michie and colleagues (Michie et al., 2011, 2014), serves as a tool to design interventions that have a high likelihood of uptake and compliance among target populations (Biesheuvel et al., 2021; Reid et al., 2022; Murphy et al., 2023). It was developed with the purpose of overcoming limitations of prior frameworks (e.g., lack of comprehensiveness, lack of coherence, no link to overarching model of behavior). At the core of the BCW is the capability, opportunity, and motivation (COM-B) framework, or behavior system. Encircling the hub are spokes consisting of 9 intervention domains, such as education, persuasion, incentivization, coercion, and training, that can be implemented to create behavioral change. The rim of the wheel contains policy categories, describing policy approaches to support behavior change, such as regulations, guidelines, and fiscal measures (Michie et al., 2011).

Using the BCW helps ensure that interventions are well-grounded in behavioral science and theory, increasing their chances of success. The use of the COM-B framework is growing in popularity in the context of livestock research (Carroll and Groarke, 2019; Surendran et al., 2023; Burrell et al., 2024). This is further illustrated by recent publications aimed to explore farm-

ers' and veterinarians' behavior toward AMS and AMU regulations (Rees et al., 2021a; Farrell et al., 2023a,b; Millar et al., 2023). By using the COM-B framework, AMS programs can draw on grounded recommendations to address factors that influence antimicrobial treatment behavior.

The aims of this qualitative study were, therefore, (1) to understand Western Canadian dairy farmers' perceptions of mastitis-related AMS; (2) to identify barriers and motivators to the adoption of milk culturing as part of Western Canadian farmers' CM treatment protocols using the COM-B model; and (3) to provide suggestions for potential interventions to support the adoption of milk culturing practices using the BCW.

MATERIALS AND METHODS

This study received ethical approval from the University of Calgary Conjoint Faculties Research Ethics Board (no. REB21-0699). The reporting of this study is in accordance with the Consolidated Criteria for Reporting Qualitative Research (COREQ) framework (Tong et al., 2007). As this research was conducted in parallel with another research project, the positionality statement and a more extensive description of the data collection process are reported elsewhere (de Jong et al., 2024b).

Data were collected through semistructured interviews with 17 dairy farmers, held over Zoom (Zoom Video Communications, San Jose, CA) and in-person, between June 2022 and July 2023. The farmers were recruited through existing connections of the researchers and through extension events. The interview guide (<https://data.mendeley.com/datasets/hv9h7k499b/1>) was designed to characterize decision-making processes for CM treatments (reported in de Jong et al., 2024b) and to inquire about mastitis-related AMS. Specifically, farmers were asked to articulate their understanding of the term AMS, what responsible use of antimicrobials means for their farm operation, and how they could improve their mastitis-related AMU. In addition, interviewees were asked their thoughts on culturing of CM milk samples, a central element of a selective CM treatment protocol and an important way to improve mastitis-related AMS (de Jong et al., 2023a).

Each interview was audio recorded and subsequently transcribed using the automated transcription program Otter.ai (version 3.116.0; Los Altos, CA). Transcripts were verified by author EdJ through comparison with the original audio recordings to confirm accuracy, and participants were assigned anonymous identification codes (P1–P17). EdJ manually applied codes to all 15 hard-copy transcripts, focusing on segments that offered meaningful insights into mastitis-related AMS and segments related to barriers and facilitators to adopting reg-

ular culturing of CM samples. Codes were transferred to the online diagramming tool draw.io (JGraph Ltd., version 29.7.9). Codes related to perceptions of AMS were inductively grouped by EdJ into themes using reflexive thematic analysis (Braun and Clarke, 2006), where themes “capture a wide range of data that are united by, and evidence, a shared idea” (Braun and Clarke, 2022, page 77). Inductive grouping was selected to highlight patterns revealed within the dataset. The codes associated with CM culturing behavior were deductively grouped, according to the domains outlined in the COM-B framework (physical capabilities, psychological capabilities, physical opportunities, social opportunities, automatic motivation, and reflective motivation), and to identify areas where intervention strategies could be developed to increase uptake of culturing. The themes were continuously refined through revisiting transcripts, discussions with the study team, and adaptations of the interview guide. EdJ discussed codes with authors JAI, AMCS, and HWB at regular intervals throughout the coding process. Theme identification was also discussed among these authors and later reviewed by all authors. This iterative process ensured rigorous data collection and analyses, resulting in a strong representation of the data. Following the analyses, the BCW was used to develop theory-based recommendations for interventions to influence behavioral change, with each recommendation connected to a component of the COM-B framework (Michie et al., 2014).

In this study, the behavior of interest was adoption of CM culturing and CM treatment decisions based on culture results—both central elements of a selective CM treatment protocol. The BCW intervention functions built the foundation for the development of recommendations to address the barriers identified by the interviewees, to align with the motivators, and to reflect an understanding of the farmers' perspectives of AMS. This approach ensures that the interventions resonate with the unique challenges and motivations encountered within the studied population (Michie et al., 2011).

RESULTS AND DISCUSSION

In total, 15 interviews were conducted with 17 farmers from British Columbia (n = 6 interviews) and Alberta (n = 9 interviews), Canada. The initial 3 interviews were conducted in-person and served as an opportunity to also elicit feedback on the interview guide. The remaining 12 interviews were conducted through Zoom. A detailed description of the participants and their farms can be found in de Jong et al. (2024b). In brief, most of the interviewees were male (88%). The farms milked an average of 176 cows, using either a parlor (53%) or an automated milking system (47%). All farms used freestall barns to

Table 1. Themes and descriptions of mastitis-related antimicrobial use based on 15 interviews with dairy farmers from Alberta (n = 9) and British Columbia (n = 17), Canada¹

Theme	Description
1. Dairy farmers' AMS decisions: Responsible in principle, unintentionally variable in practice 2. Upstream solutions: Prevention, herd health, and good management as foundations of judicious AMS 3. Need for consistent engagement of the veterinarian as an external advisor on AMS 4. Behavioral change around AMS: Generational differences, internal drive, external support	Although interested in reducing AMU on their farms, farmers unintentionally engaged in practices they described as unfavorable to AMS. Improving mastitis-related AMS should emphasize reducing disease incidence. Prevention was highlighted as a key factor contributing to AMS, specifically cleanliness and good milking practices to improve udder health. Involvement and support of the veterinarian in AMU was recognized as an indication of good AMS. Impactful change requires a communal effort. Effort is required to disseminate knowledge and to reach older generations and those with less intrinsic motivation.

¹AMS = antimicrobial stewardship; AMU = antimicrobial use.

house the cows. Interviewees were a combination of farm owners and herd managers, and all participants were involved in CM treatment decisions.

Dairy Farmers' Perceptions of Mastitis-Related AMS

Four themes were identified regarding the farmers' perceptions of mastitis-related AMS: (1) dairy farmers' AMS decisions: responsible in principle, unintentionally variable in practice; (2) upstream solutions: prevention, herd health, and good management as foundations of judicious AMS; (3) the need for consistent engagement of the veterinarian as an external advisor on AMS; and (4) behavioral change around AMS: generational differences, internal drive, and external support (Table 1).

Theme 1. Dairy Farmers' AMS Decisions: Responsible in Principle, Unintentionally Variable in Practice

Interviewees expressed a strong conviction in practicing good AMS by treating only when deemed necessary, a perspective that aligns with findings reported by others (Cobo-Angel et al., 2021; Wemette et al., 2021). Their approach was driven by ethical considerations, cost concerns, and belief in using antibiotics as needed for sick animals. However, in some cases, although farmers expressed understanding of and intention to practice judicious AMS, current practices did not always align. One farmer emphasized avoiding unnecessary AMU while prioritizing antimicrobial treatment of cases where it is needed: "You don't just stick drugs in them because we're like, 'Let's just drug them.' We'd all prefer not to do that. But there are some cows that obviously need a little bit more help than others" (P17). Another farmer stated, "We're not going to use more antibiotics than we have to. Not only from an ethical point of view, but also from a cost [perspective]" (P10). Dairy farmers also expressed an ethical responsibility to safeguard their animals' health and well-being in

their decisions to treat cows with antibiotics. The prioritization of animal welfare is commonly mentioned among farmers and veterinarians when asked about their hesitations for reducing antibiotic use (Poizat et al., 2017; Scherpenzeel et al., 2018; Cobo-Angel et al., 2021; Wemette et al., 2021; Ida et al., 2023).

Farmers were generally aware of practices that are considered "good" and "bad" regarding the responsible use of antibiotics. Some of the unfavorable practices they noted included extended duration of treatment, improper dosage, incomplete treatment, and changing antimicrobials without veterinary consultation when one stopped working. For example: "I don't think they like it any more that you do that. Like, combining antibiotics" (P10). These indicate a reasonable understanding of AMS principles. Interviewees also listed other practices that assist mastitis-related AMS, such as strategic culling of cows with frequent health issues, decreasing disease incidence through vaccination programs, and enhancing herd health through selective breeding, similar to focus areas mentioned by Eastern Canadian dairy farmers (Cobo-Angel et al., 2021). For example: "It can be a good bull, but this [mastitis index] is too high for us" (P6).

However, simultaneously, farmers unintentionally engaged in practices that impeded their intentions to act in accordance with optimal AMS guidelines. These became clear upon closer examination of the voiced CM treatment decisions. Farmers were either unaware that these practices were suboptimal or justified them, often due to a lack of knowledge. For example, regarding extending intramammary treatments: "I thought I was being more ethically responsible by treating them for longer if I wasn't seeing improvement. I thought that we have a better chance of killing the bacteria and there's less a chance of antibacterial resistance or whatever" (P10). Extending treatment is considered extralabel AMU, which may be pursued in some cases but requires a valid veterinarian–client–patient relationship and direct guidance from the veterinarian. Another farmer shared that not all cows with mastitis receive antimicrobial

treatment, but then stated, “There are definitely times when I haven’t treated or there [have been times] if they have high somatic cell count after a milk test, I’ll just give them Excede [ceftiofur crystalline free acid; third-generation cephalosporin antibiotic], [and] kick them out. Because it’s not a [milk] tank withholding drug. But it seems to just help what ails them” (P17).

Theme 2. Upstream Solutions: Prevention, Herd Health, and Good Management as Foundations of Judicious AMS

Farmers emphasized prevention and low incidence of disease as upstream drivers of good AMS, similar to reports from Norwegian farmers, who indicated that achieving good herd health with minimal antibiotics gave a sense of satisfaction (Hektoen, 2004). Farmers demonstrated awareness of the need to safeguard antimicrobials that are critically important for human health, as well as the role that milk culturing can play in improving AMS. Farmers in this sample also emphasized that antibiotics should not compensate for poor management practices, further reinforcing prevention as the cornerstone to effective stewardship. Farmers emphasized that on farms where large quantities of antimicrobials were administered, reducing disease incidence should be the priority: “To treat every cow with antibiotics isn’t responsible use because upstream from that, not every cow should have mastitis” (P13). Some farmers demonstrated extensive knowledge of the debate about important antimicrobials. According to one farmer, “Cephalosporin 4 [fourth generation] are the next drugs for humans. Trying to minimize our usage of them in food production, through production animals, would probably benefit [us] as an industry if we were away from that” (P14). Another farmer stated, “I think, someday we probably should [culture milk]. Culture would even further reduce antibiotic use. That would be nice” (P15).

Prevention was highlighted as a key factor contributing to AMS; specifically, cleanliness and good milking practices to improve udder health were mentioned. This is similar to key areas identified in interviews with veterinarians (Farrell et al., 2023b) and other dairy farmers (Cobo-Angel et al., 2021; Wemette et al., 2021). Interviewees raised concerns about absence of prevention in the context of AMS, specifically during interviews in which prevention was not explicitly mentioned by the interviewer. Therefore, it is important for veterinarians and other dairy industry service providers to provide support for improving udder health on dairy farms before or in conjunction with proposing potential changes to the farm’s AMU. One farmer shared, “I think the number-one thing that dairy farmers can do is the preventative measures, so they don’t have to use the drugs, they don’t

get the mastitis” (P17). Another stated, “I’m going to do more to prevent cases, to prevent using” (P14).

On farms with low disease incidence, improving AMS was not deemed a top priority, as articulated by one farmer: “It [AMR] is not anything that we are too concerned about. And that is probably more so [because of] the fact that we might only treat a mastitis cow once a month” (P5), which accords with interviews among sheep, dairy, and beef producers in the UK (Golding et al., 2019). Similarly, farmers who perceived themselves to be using minimal amounts of pharmaceuticals were not concerned about AMR, as voiced by one farmer: “Seeing the amount of medicine we use, I am not worried about it [AMR]” (P11). A consensus that antibiotics should not serve as a solution to inadequate herd management was identified, in line with findings from Ida et al. (2023). One participant explained, “If you’re treating too many cows, maybe you should be looking at something else. Is [there] something you’re doing wrong on the farm? Something you could be doing better?” (P15) Another farmer makes the connection to environmental management, specifically to stall cleanliness: “I was talking to someone the other day who had a higher somatic cell count. But they have deep bedding, and it’s hard to keep their stalls clean” (P7).

Theme 3. Need for Consistent Engagement of the Veterinarian as an External Advisor on AMS

Veterinarians serve as advisors to farmers, especially in the context of AMS (among others, Lam et al., 2011; Jones et al., 2015; Swinkels et al., 2015; Fischer et al., 2019; Cobo-Angel et al., 2021; Ida et al., 2023), and can help mitigate concerns surrounding adoption of new practices (Brunt et al., 2025). In line with this, the interviewed farmers placed a high degree of responsibility on veterinarians to identify and address suboptimal practices, flag excessive AMU, and offer insights into disease challenges on farm. Farmers value veterinarians’ external perspective, seeing them as trusted outsiders who can identify issues that may be overlooked by those involved in the day-to-day activities of the farm. One farmer said, “Our vets? They’re an information source. They are much more up-to-date on the new drugs available. [And they know how] we can utilize different cost-effective drugs that are not necessary [for human medicine], such as your penicillins. As opposed to trying to utilize these cephalosporins, and more modern drugs” (P14). The benefit of the veterinarian’s role as an outsider was specifically highlighted by this farmer: “I’m in it every day. Sometimes you don’t see things changing because you’re so in it. If you are on the outside, like the vet does, he can see certain things [. . .]. That’s definitely something I’m expecting from the vet. Because he’s an outsider. He

helps us, he's supporting us. So, he knows us, but he's not involved daily" (P8). Although "outsiders" are not universally trusted in all settings, farmers in this context see value in the external perspective of the veterinarian, and the insight and guidance that comes with occupying a position external to the farm.

The involvement of the veterinarian, especially in the development of protocols, has been shown to instill confidence in farmers regarding their AMS (Fischer et al., 2019; Cobo-Angel et al., 2021), a point that also was identified from the interviews, as illustrated by one farmer: "If you follow [the protocol developed with the veterinarian] as recommended, you don't go off-label, and you don't go over label, you do your proper dosages. I can't say there'll be too much [antimicrobial resistance] to worry about" (P15).

However, a wide range of factors are known to influence veterinarians' knowledge, attitudes, and awareness toward AMS (Farrell et al., 2021), potentially leading to unmet expectations by farmers. One farmer highlighted that the age, attitude, and longstanding professional relationship with their veterinarian affect their dynamics which, in their case, reduced opportunities for AMS: "Our vet on-site, we've had him for, like, a zillion years. He's going to retire any moment. And a lot of the times they [the co-owners] don't agree with his ideas. But we use him because we've used him for so long. I think once we change vets, a lot is going to actually change in protocols. You know, he [our older vet] is set. And it's always how we've done it. [We] talked to some of the newer vets. They're like, 'Oh this is what the research . . .' and they all have their numbers. 'They found if you use this or that . . .' I think change comes with the younger generation" (P17). In addition, industry-wide improvements in cow health (e.g., transition cow management, genetics, and nutrition) have influenced the role of the veterinarian and reduced visit frequency, which subsequently reduces opportunities to identify problem areas for AMS: "You used to have the vet out a few times a year for heavy calving, but those days are over [. . .]. Often times, I phone them for advice. But it's not like they come here and see things anymore" (P7).

From the interviews, it became evident that veterinarians emphasized the importance of AMR awareness during their interactions with farmers: "Our vets pretty much have said over the years, 'If you use Cefa-Lak [intramammary cephalosporin sodium] on all your mastitis cows year after year, it's probably not going to work. Switch up your drug every once in a while with something else'" (P5). However, the interviews also revealed instances of veterinary advice opposing prudent AMU. For instance, one farmer reported that their veterinarian advocated for extended treatment, and multiple farmers expressed that their veterinarians endorsed blanket dry cow therapy.

Veterinarians and farmers in the UK reported variability in veterinarians' engagement with "good" AMS practices (Golding et al., 2019). Several factors have been hypothesized to influence this engagement, including the challenge of integrating AMS into everyday farming practice, cognitive dissonance among farmers between knowledge and behavior (Golding et al., 2019), and pressure from clients and other influential non-veterinary farm advisors (Speksnijder et al., 2015).

Theme 4. Behavioral Change Around AMS: Generational Differences, Internal Drive, External Support

Adoption of new AMS practices is driven by internal motivation, influenced by generational differences, and needs to be supported by external actors. As in many other agricultural professions, older generations of dairy farmers are more resistant to change (Brown et al., 2019), which was also voiced in the interviews: "They [older dairy farmers] have a hard time changing their attitude, changing their ways, 'We've always done it this way'" (P17). However, participants noted that willingness to adopt change may be influenced not only by age but also by intrinsic motivation. For instance, farmers who are motivated by a genuine desire to acquire the right knowledge and skills are likely more inclined to successfully implement change on the farm. One farmer illustrated, "If the farmer doesn't go to your presentation, or they don't ask the vet questions, then you can't really do a whole lot either. It's up to the farmer to be involved and proactive and to learn. Unless you go knocking on every farmer's door; but even then, they won't always be receptive to you" (P7).

Although initiatives such as proAction (<https://www.dairyfarmers.ca/proaction>), which includes tracking treatment administration, are in place to support farmers in achieving AMS goals, their implementation and practical realities have been met with frustration by some farmers. The long-winded extensive documentation process associated with proAction was mentioned by interviewees, alongside a fear that more paperwork might be demanded of them to reach future targets surrounding AMU, which would take away time spent observing and directly caring for cows: "With proAction [validation visit], last time I said, 'I hope they enjoy me sitting behind the computer, because I'd rather spend my time with the cows.' It is for the people that manage a little less that do need that. There are certain things about proAction that are really good [. . .]. But we got to watch out for that you push too far" (P6).

The influence of other dairy farmers, society, and farm advisors on farmers' AMS beliefs and practices should not be overlooked (Swinkels et al., 2015; Brunt et al.,

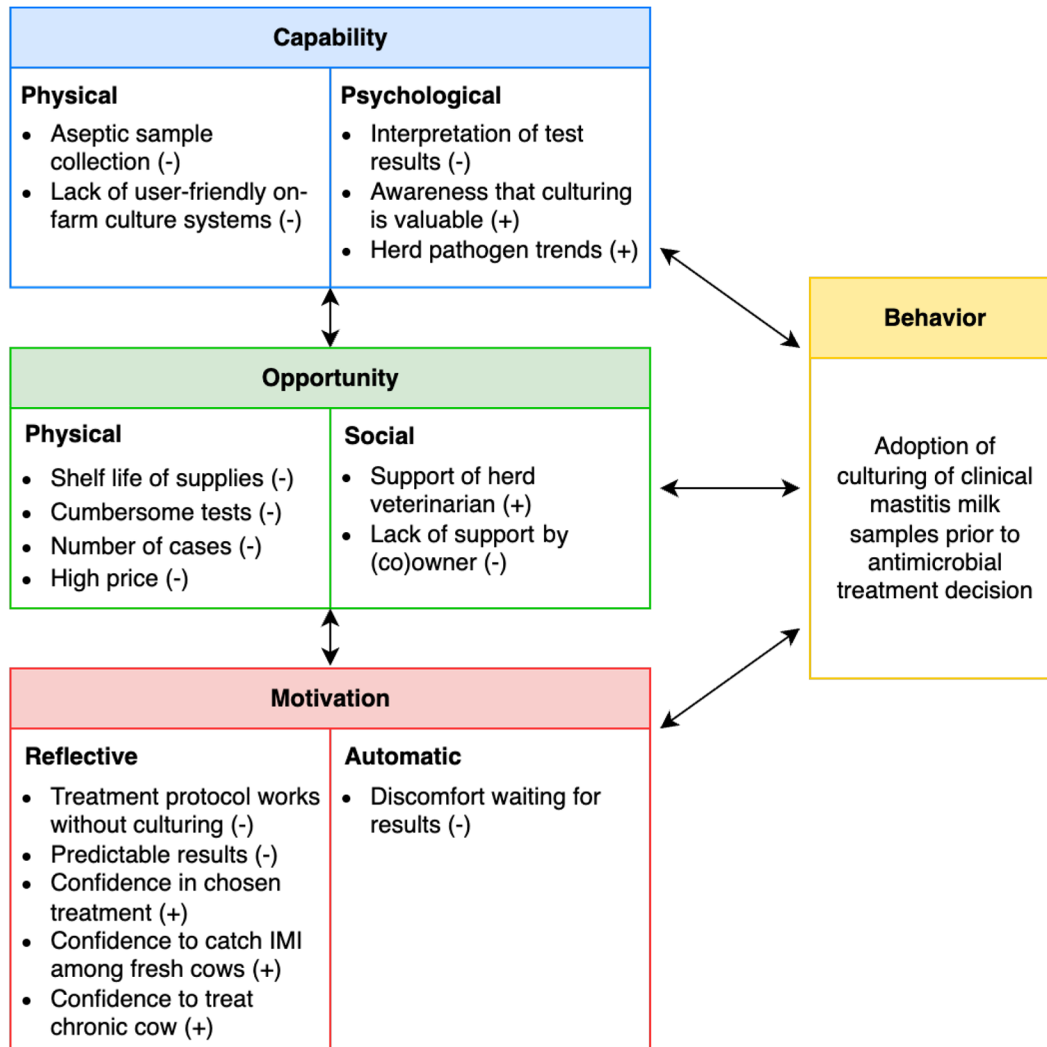


Figure 1. Barriers (–) and facilitators (+) for the adoption of culturing of clinical mastitis cases, based on interviews with 15 farmers from British Columbia and Alberta, Canada. Grouped according to the COM-B framework (Michie et al., 2011).

2025). Empowering industry service providers and researchers could further aid in disseminating responsible AMU knowledge to both veterinarians and farmers. The importance of these different actors, all working toward a similar goal, was underscored by one farmer as, “It takes a village to make a change” (P17).

Barriers and Facilitators to Culturing of CM Samples, and Suggested Interventions

In all, 6 facilitators and 10 barriers were identified regarding the adoption of milk culturing in farmers’ CM treatment decision process, which were classified according to the COM-B framework (Figure 1). Using the BCW framework, 10 interventions were designed to promote the target behavior (CM cases cultured and the culture results used to inform treatment decisions regarding an-

timicrobial CM treatments). The interventions aimed to mitigate the known barriers, support the facilitators, and be reflective of the farmers’ views on AMS (Table 2).

Physical Capability. Many of the farmers who had experience with culturing CM cases indicated that contamination was often reported, discouraging continuation of regular culturing of CM samples: “And then they’d come back as contaminated. So, you go through all that effort and cost, and then there’s no real data” (P7). This indicates a challenge with aseptic milk collection, which is a crucial skill in the successful adoption of milk culturing practices. The responsibility for providing proper training primarily falls upon the veterinary clinic or laboratory; however, industry service providers can also be engaged. Especially on farms with only a few CM cases a month, sustaining the skill of aseptic sampling presents a challenge, and

Table 2. Suggested interventions designed with the Behavior Change Wheel (Michie et al., 2011)¹

Intervention	Intervention function	Domain
Frequent training for farm personnel, focused on aseptic milk sampling to prevent contamination	Training	Capability
Include information in culture reports regarding interpretation of results, especially the interpretation of “negative” test results	Education	Capability
Frequent check-in from veterinarian to ensure proper use of on-farm culture and interpretation of results	Education	Capability
Increase shelf life, improve ease of use, and reduce costs per plate of on-farm culturing systems	Environmental restructure	Opportunity
Offer udder health packages at veterinary clinic (covering costs of culturing for individual CM cases or to develop a herd pathogen profile)	Enablement	Opportunity
Promote milk culturing as part of selective treatment protocols among different farm personnel through various information sources, including farm magazines and the herd veterinarian	Persuasion	Opportunity
Hold demonstration sessions at farms with milk culturing as part of treatment protocol to normalize culturing	Modeling	Motivation
Address discomfort and anxiety associated with waiting for culture results before initiating treatment	Modeling	Motivation
Highlight increased confidence of chosen treatment approach through demonstration sessions and peer study groups	Modeling	Motivation
Benchmark uptake of milk culturing and clinical mastitis-related AMU to normalize culturing	Incentivization	Motivation
Increase problem awareness: goal of culturing is not to improve treatment outcomes but to minimize AMU	Education	Motivation

¹CM = clinical mastitis; AMU = antimicrobial use.

frequent revisiting of the training will be important (de Jong et al., 2023a).

Psychological Capability. Similarly, “no growth” or “culture-negative” results were viewed as undesirable and often mentioned as a reason to stop culturing CM samples, as illustrated by one farmer: “But alas, for 3 [CM samples], it was no growth of them. So, then it’s hard to know how to treat them” (P6). A negative test result of CM culturing is common (Oliveira et al., 2013) and indicates that no bacterial thresholds were met, often a result of absence of bacteria in the udder at the time of sampling. The correct treatment decision for cases in which there is no growth on the plate should be not to administer antibiotics, although this was not always clear to the farmers. This points to a challenge experienced by farmers regarding the proper interpretation of culture results. To enhance understanding and educate farmers, clear and easy-to-interpret explanations should be included in diagnostic reports when culturing is performed at a laboratory. Alternatively, when on-farm culturing systems are employed, veterinarians or extension workers should check in regularly to ensure proper understanding of the results.

Physical Opportunity. Farmers mentioned that on-farm culture of CM cases would be more appealing when easier-to-use systems are developed. Examples provided by farmers included cow-side tests similar to pH strips and snap tests used to detect antibiotic residue in milk. The already available Mastitis4 test, a milk PCR test offered by Lactanet (DHI; Guelph, ON, Canada) that identifies *Staphylococcus aureus*, *Streptococcus agalactiae*, *Mycoplasma bovis*, and *Prototheca* spp., was met with positive responses. One farmer explained, “It’s the quickest way to get tested; the sample is already taken. Otherwise, I have to wait to get the [somatic cell count] results back from DHI, then I have to take samples and send them to a

lab” (P8). In addition, shelf life of on-farm culturing systems often hindered their use, particularly on farms with low CM incidence, and especially considering the costs of each culture, as illustrated by one farmer: “The cost of those cultures too. That 25 bucks for 1 culture, and then they get old and dry out. So, then it’s kind of a waste of money again. So I stopped doing it” (P7). On farms with few CM cases, on-farm culturing systems were also less convenient to implement due to the costs per case: “Then I stopped doing that [culture]. Partly because I don’t have a lot of cases” (P7).

Addressing the limited shelf life of products and more user-friendly tests largely falls within the scope of manufacturers, although some of these factors are outside their control. However, the barrier of cost per culture, also reported by others (Friedman et al., 2007), can be mitigated without the manufacturer’s involvement. Cost per sample could be integrated into more comprehensive udder health programs often provided by veterinary clinics. Such a strategy might eliminate the barrier for farmers deciding to culture individual CM cases. Costs could be lowered by only offering to culture CM cases occasionally, with the intent to create a herd pathogen profile. Such a profile can help predict the cause of future cases, tailoring treatments.

Social Opportunity. Similar to other studies that have demonstrated the importance of a trusted relationship with the veterinarian (Brunt et al., 2025), the herd veterinarian was often mentioned as pivotal in the adoption of milk culturing in CM treatment protocols. One farmer in particular, who had been culturing CM samples for over 10 years, explained that his veterinarian pushed for this: “It was probably the vet that pushed it the most. We can do this. You probably get better results, knowing what mastitis you’re actually treating. But there are different antimicrobials that you can use. So, you might

as well treat it with the right one rather than just taking a shot in the dark” (P16). This role extends not only to farms initially hesitant to adopt CM culturing but also to those facing internal divisions, where some individuals advocate for its adoption and others remain skeptical. For example, the following farmer worked with his father who was not supportive of culturing: “I would have to [personally] take [a] milk sample and send it for culture because my dad, he’s kind of an old-time farmer. And, yeah, he wouldn’t want to do that” (P9).

Effective communication strategies are key, especially in contexts where a community of farmers hold varying perspectives toward information (Jansen et al., 2010). Jansen et al. (2010) described that farmers exhibit a spectrum of openness or resistance to new information, as well as varying degrees of trust and distrust of information sources. It is essential to recognize that a farmer’s reluctance to change does not necessarily reflect a lack of value placed on their udder health program; rather, it calls for a tailored approach to engagement and communication. Some farmers also rely heavily on farm magazines as a trusted source of information (Jansen et al., 2010; Cobo-Angel et al., 2021). Leveraging this preference could present a promising avenue for further promoting the adoption of culturing in managing CM cases.

Reflective Motivation. Farmers also mentioned that, when their current treatment protocol worked, there was less desire or need to include additional diagnostics such as culturing of CM cases before making treatment decisions. One farmer said, “We never really have [a need]. Some cases might take longer for them to clear up, and we sample those cases to see what we’re dealing with” (P15). Similarly, during discussions with the interviewees on strategies to enhance AMS, mentions of culturing or improved diagnostics were relatively rare. It is important to address this lack of awareness, as the goal of culturing is not to improve treatment outcomes but to reduce unnecessary AMU and to limit AMR. Additionally, although culturing involves added diagnostic costs, it may result in overall savings in treatment-related expenses.

Another identified barrier was “predictable results.” When cultures of different CM cases yielded the same results, farmers were discouraged and unmotivated to continue culturing, an issue also noticed by Dutch veterinarians (Speksnijder et al., 2015). One interviewee shared, “We did do it [culture] for a while . . . then it just felt like we were throwing money away.” In these instances, the option to culture only a portion of cases to inform herd pathogen trends could be highlighted.

Some farmers, however, did express that culture results could provide confidence in their decision-making. For example, in decisions to withhold treatment of chronic mastitis cows: “I did it mainly on those

[chronic] cows to see whether they had actually bugs in the quarters or if it was just inflammation causing flakes to show up” (P1). Culturing was also used to mitigate uncertainty of treatment decisions for nonsevere CM cases: “Some cases might take longer for them to clear up, and we sample those cases to see what we’re dealing with” (P15). Interviewees identified that culturing could also be helpful to identify IMI in fresh cows and those with high SCC on DHI tests.

Automatic Motivation. Automatic motivators, which include impulses, as opposed to reflective motivators, which include processes of choice and intention (Michie et al., 2011), were described by the interviewees. Automatic motivators, for example, the inclination to immediately treat, were weighed against reflective motivators, specifically evidence-based decisions regarding the typical time to results of milk cultures, which corresponds with experiences reported by Irish dairy veterinarians (Farrell et al., 2023b). More specifically, receiving results 24 to 48 h after sending in a sample was undesirable, as it conflicted with the desire to treat a case right away to prevent deterioration of the cow’s condition: “If we would have waited a day for that information to come back, then we would have been a day behind to treat the cow” (P1). Interventions such as demonstration sessions and peer study groups, where farmers share experiences, could be excellent tools to minimize the discomfort associated with the wait for culture results. Such groups have demonstrated their ability to shape perceptions in the context of Johne’s disease control (Roche et al., 2015). For example, farmers who have adopted culturing could share their experiences regarding treatment outcomes when waiting 24 to 48 h for culture results, directly addressing others’ concerns about waiting for results. Additionally, research demonstrating that selective treatment of CM based on on-farm culture results shows no long-term differences in health and production outcomes (Lago et al., 2011) should be better communicated by researchers and extension specialists.

Additional Notes on Interventions. Each of these recommendations assigns a major role to the herd veterinarian, not only to provide information to the farmer but also to learn about farmers’ immediate and long-term motivational drivers (Bard et al., 2019), to offer to share ideas about culturing CM samples if in line with the farmer’s long-term goals, to educate farmers about the interpretation of tests, to train farmers regarding aseptic sampling, and to mitigate discomfort associated with waiting for results. This role may not be suited to every herd veterinarian, especially considering the known uncertainty in farmers’ responses when veterinarians propose changes in treatment protocols (Speksnijder et al.,

2015; Golding et al., 2019). Veterinarians might consider enlisting udder health experts or veterinary technicians to carry out these duties. Using a roster or a schedule to systematically review management practices throughout the year could further help address issues on dairy farms; for example, the 4-mo cycle designed by Ruegg (2022) to address CM AMS. Providing a tailored approach—for example, one that acknowledges the specific barriers that an individual community is facing—has been proven to be effective in various settings aimed at changing behavior (Noar et al., 2007).

Limitations

The strength of obtaining data through interviews, as opposed to surveys, is the depth, nuance, and accuracy of information gleaned. Although the semistructured interview allowed for probing, the interviews were not conducted within the context of ethnographic fieldwork. As such, broader contexts, personal and cultural identity, social influences other than the herd veterinarian, and other aspects known to influence perception of AMU among dairy farmers (Swinkels et al., 2015; Ida et al., 2023) were not specifically explored. The majority of interviews were conducted remotely. Although this may alter the researcher–participant relationship (Ritter et al., 2023), this was not believed to significantly influence the interviews, which is in line with findings from Roberts and colleagues (2025). Through a scoping review, they found that remote qualitative data collection provides similar quality and depth to in-person qualitative methods. The interviewed farmers were largely part of the personal network of the research team, and a level of rapport was already established before the interviews. This likely fostered greater trust between participants and interviewers, resulting in more open and authentic responses. However, we recognize that not all potential limitations of remote data collection can be mitigated by prior relationships with participants. It is possible that nonverbal cues were missed and that sensitive topics were avoided by the participant (Roberts et al., 2025). The personal relationships of the research team with the farmers could have invited socially desirable answers, despite efforts made during the interview to probe for genuine responses and highlight the value placed on the farmers' own perspectives. Additionally, although generalizability is not a primary objective of most qualitative research, it is possible that the sample represents only a portion of farmers' perspectives, which may not be generalizable to all farmers in Canada. With this in mind, a survey among a larger farmer population could be a useful tool to validate and tailor the recommendations made in this manuscript to the broader Canadian dairy industry.

CONCLUSIONS

In this study, we explored Western Canadian dairy farmers' perceptions of mastitis-related AMS and adoption of milk culturing. Four themes were identified regarding farmers' perceptions of mastitis-related AMS. Despite their intentions to minimize AMU, the farmers in our study unintentionally engaged in practices they described as unfavorable to AMS (Theme 1). Farmers equated low incidence of CM with good AMS (Theme 2) and emphasized the importance of the involvement and, thereby, approval of the veterinarian in decision-making (Theme 3). Participants also noted that fostering change is a communal effort that takes time and patience (Theme 4). Additionally, we identified barriers and facilitators to the uptake of milk culturing, a key component to a selective treatment approach, using the COM-B framework. We used these insights to develop potential interventions that support AMS using the BCW. Implementing these interventions could lead to increased uptake of selective CM treatment protocols, facilitating long-term improvement of on-farm AMS practices.

NOTES

This study was funded by Canada's Natural Sciences and Engineering Research Council (NSERC; Ottawa, ON, Canada) Industrial Research Chair Program granted to author Herman Barkema, with industry contributions from Alberta Milk (Edmonton, AB, Canada), Dairy Farmers of Canada (Ottawa, ON, Canada), Dairy Farmers of Manitoba (Winnipeg, MB, Canada), British Columbia Dairy Association (Burnaby, BC, Canada), WestGen Endowment Fund (Abbotsford, BC, Canada), Lactanet (Guelph, ON, Canada), SaskMilk (Regina, SK, Canada), and MSD Animal Health (Boxmeer, the Netherlands). Data collection forms, interview guides, analytic codes, and any other materials used are available from the corresponding author upon request. This study received ethical approval from the University of Calgary Conjoint Faculties Research Ethics Board (no. REB21-0699). The reporting of this study is in accordance with the Consolidated Criteria for Reporting Qualitative Research (COREQ) framework (Tong et al., 2007). As this research was conducted in parallel with another research project, the positionality statement and a more extensive description of the data collection process are reported elsewhere (de Jong et al., 2024b). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AMR = antimicrobial resistance; AMS = antimicrobial stewardship; AMU = antimicrobial use; BCW = behavior change wheel; CM = clinical mastitis; COM-B = capability, opportunity,

and motivation behavioral change framework; P1–P17 = participant identification codes.

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ORCIDS

Jennifer A. Ida,  <https://orcid.org/0000-0003-0703-0870>
 Ellen de Jong,  <https://orcid.org/0000-0002-4198-7898>
 Anne-Marieke C. Smid,  <https://orcid.org/0000-0001-9810-4032>
 Kristen K. Reyher,  <https://orcid.org/0000-0003-4912-5026>
 David F. Kelton,  <https://orcid.org/0000-0001-9606-7602>
 Caroline Ritter,  <https://orcid.org/0000-0001-7349-5241>
 Herman W. Barkema  <https://orcid.org/0000-0002-9678-8378>